

20장 과제

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□ apt update와 apt upgrade 명령의 차이를 설명하라.

apt update는 패키지 목록을 갱신하는 것이고, apt upgrade는 실제로 패키지를 업그레이드하는 과정입니다. 보통 패키지를 업그레이드하기 전에 apt update를 먼저 실행해야 합니다.

□ apt update와 apt upgrade 명령을 이용하여 현재 설치된 모든 패키지를 최신상태로 업데이트하라. 설치관련 명령어는 관리자권한이 필요하므로 sudo를 앞에 추가하고실행하라.

```
linux@localhost:~$ sudo apt update
[sudo] password for linux:
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [1848 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2066 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [299 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [358 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [17.9 kB]
```

```
Linux@localhost:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done

The following packages were automatically installed and are no longer required:
 htop libnl-3-200 libnl-genl-3-200
Use 'sudo apt autoremove' to remove them.

The following NEW packages will be installed:
 adwaita-icon-theme at-spi2-core bc cloud-guest-utils cloud-init dconf-gsettings-backend dconf-service
 eatmydata fdisk fontconfig fontconfig-config fonts-dejavu-core gdisk gsettings-desktop-schemas
 gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme landscape-client landscape-common
 libatk-bridge2.0-0 libatk1.0-0 libatk1.0-data libatspi2.0-0 libavahi-client3 libavahi-common-data
 libavahi-common3 libcairo-gobject2 libcairo2 libcolor2 lib cups2 libdatatr1 libdconf1 libdeflate0
 libdrm-amdgpu1 libdrm-intel1 libdrm-nouveau2 libdrm-radeon1 libeatmydata1 libegl-mesa0 libegl1 libepoxy0
 libfdisk1 libfontconfig1 libfreetype6 libgbm1 libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin
 libgdk-pixbuf2.0-common libgl1 libgl1-amber-dri libgl1-mesa-dri libglapi-mesa libglvnd0 libglx-mesa0 libglx0
 libgraphite2-3 libgtk-3-0 libgtk-3-bin libgtk-3-common libharfbuzz0b libjbig0 libjpeg-turbo8 libjpeg8
 liblmdb2 liblvm2-2 liblzma5 libnss3-1-0 libnsscore1-0 libnsscore2-1-0 libnsscore3-1-0 libnsscore4-1-0 libnsscore5-1-0
```

```
linux@localhost:~$ sudo apt full-upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  htop libnl-3-200 libnl-genl-3-200
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
linux@localhost:~$
```

```
linux@localhost:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
linux@localhost:~$
```

□ apt search, apt show 명령어의 차이를 설명하라.

apt search는 키워드를 기반으로 관련 패키지를 찾는 검색 용도로 사용됩니다. 패키지의 이름과 간단한 설명만을 출력합니다.

apt show는 특정 패키지에 대해 상세 정보를 조회하는 데 사용하며, 해당 패키지의 버전, 의존성, 유지보수자 등 구체적인 정보를 제공합니다.

□ apt remove, apt purge, apt autoremove 명령어의 차이를 설명하라.

apt remove: 패키지를 제거하지만, 설정 파일은 남겨둡니다.

apt purge: 패키지과 설정 파일을 모두 제거.

apt autoremove: 더 이상 필요하지 않은 의존성 패키지들을 자동으로 제거.

□ 현재 어떤 패키지가 설치되어 있는지를 확인하는 방법을 쓰시오

```
linux@localhost:~$ dpkg --get-selections
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name                    Version                Architecture Description
=====
ii adduser                   3.118ubuntu5           all          add and remove users and groups
ii adwaita-icon-theme       41.0-1ubuntu1          all          default icon theme of GNOME (small subset)
ii apparmor                 3.0.4-2ubuntu2.4       amd64       user-space parser utility for AppArmor
ii apport                   2.20.11-0ubuntu82.6    all          automatically generate crash reports for debugging
ii apport-symptoms          0.24                   all          symptom scripts for apport
ii apt                      2.4.13                 amd64       commandline package manager
ii apt-utils                2.4.13                 amd64       package management related utility programs
ii at-spi2-core              2.44.0-3               amd64       Assistive Technology Service Provider Interface (dbus core)
ii base-files                3.5.52build1           amd64       Debian base system miscellaneous files
ii base-passwd               3.5.52build1           amd64       Debian base system master password and group files
ii bash                      5.1-6ubuntu1.1         amd64       GNU Bourne Again SHell
ii bash-completion          1:2.11-5ubuntu1        all          programmable completion for the bash shell
ii bc                        1.07.1-3build1         amd64       GNU bc arbitrary precision calculator language
ii bind9-dnswtills           1:9.18.28-0ubuntu0.22.04.1 amd64       Clients provided with BIND 9
ii bind9-host                1:9.18.28-0ubuntu0.22.04.1 amd64       DNS Lookup Utility
ii bind9-ls:amd64            1:9.18.28-0ubuntu0.22.04.1 amd64       Shared Libraries used by BIND 9
ii binutils                  2.38-4ubuntu2.6        amd64       GNU assembler, linker and binary utilities
ii binutils-common:amd64     2.38-4ubuntu2.6        amd64       Common files for the GNU assembler, linker and binary utilities
ii binutils-x86-64-linux-gnu 2.38-4ubuntu2.6        amd64       GNU binary utilities, for x86-64-linux-gnu target
ii bsdxtrautils              2.37.2-4ubuntu3.4      amd64       extra utilities from 4.4BSD-Lite
ii bsduutils                 1:2.37.2-4ubuntu3.4    amd64       basic utilities from 4.4BSD-Lite
```

```
linux@localhost:~$ apt list --installed
Listing... Done
adduser/jammy,now 3.118ubuntu5 all [installed,automatic]
adwaita-icon-theme/jammy,now 41.0-1ubuntu1 all [installed,automatic]
apparmor/jammy-updates,now 3.0.4-2ubuntu2.4 amd64 [installed,automatic]
apport/jammy,now 2.20.11-0ubuntu82.6 all [installed,automatic]
apport-symptoms/jammy,now 0.24 all [installed,automatic]
apport/jammy-updates,now 2.20.11-0ubuntu82.6 all [installed,automatic]
apt-utils/jammy-updates,now 2.4.13 amd64 [installed,automatic]
apt/jammy-updates,now 2.4.13 amd64 [installed,automatic]
at-spi2-core/jammy,now 2.44.0-3 amd64 [installed,automatic]
base-files/jammy-updates,now 3.5.52build1 amd64 [installed]
base-passwd/jammy,now 3.5.52build1 amd64 [installed]
bash-completion/jammy,now 1:2.11-5ubuntu1 all [installed,automatic]
bash/jammy-updates,now 5.1-6ubuntu1.1 amd64 [installed]
bc/jammy,now 1.07.1-3build1 amd64 [installed,automatic]
bind9-dnswtills/jammy-updates,now 1:9.18.28-0ubuntu0.22.04.1 amd64 [installed,automatic]
bind9-host/jammy-updates,now 1:9.18.28-0ubuntu0.22.04.1 amd64 [installed,automatic]
bind9-ls:amd64/jammy-updates,now 1:9.18.28-0ubuntu0.22.04.1 amd64 [installed,automatic]
binutils-common/jammy-updates,now 2.38-4ubuntu2.6 amd64 [installed,automatic]
binutils-x86-64-linux-gnu/jammy-updates,now 2.38-4ubuntu2.6 amd64 [installed,automatic]
binutils/jammy-updates,now 2.38-4ubuntu2.6 amd64 [installed,automatic]
bsdxtrautils/jammy-updates,now 2.37.2-4ubuntu3.4 amd64 [installed,automatic]
bsduutils/jammy-updates,now 1:2.37.2-4ubuntu3.4 amd64 [installed,automatic]
```

□ 패키지 gcc를 설치하고자 한다. 다음순서로 명령어를 실행하라.

□ 패키지 gcc의 자세한 정보를 출력하라.

```
linux@localhost:~$ apt show gcc
Package: gcc
Version: 4:11.2.0-1ubuntu1
Priority: optional
Build-Essential: yes
Section: devel
Source: gcc-defaults (1.193ubuntu1)
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Debian GCC Maintainers <debian-gcc@lists.debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 51.2 kB
Provides: c-compiler, gcc-x86-64-linux-gnu (= 4:11.2.0-1ubuntu1)
Depends: cpp (= 4:11.2.0-1ubuntu1), gcc-11 (>= 11.2.0-1~)
Recommends: libc6-dev | libc-dev
Suggests: gcc-multilib, make, manpages-dev, autoconf, automake, libtool, flex, bison, gdb, gcc-doc
Conflicts: gcc-doc (<< 1:2.95.3)
Task: ubuntustudio-video, ubuntustudio-publishing, ubuntu-mate-core, ubuntu-mate-desktop
Download-Size: 5112 B
APT-Sources: http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages
Description: GNU C compiler
 This is the GNU C compiler, a fairly portable optimizing compiler for C.
.
This is a dependency package providing the default GNU C compiler.
```

```
linux@localhost:~$
```

□ 패키지 gcc의 설치 여부를 확인하라.

```
linux@localhost:~$ dpkg -l gcc
dpkg-query: no packages found matching gcc
linux@localhost:~$
```

□ 패키지 gcc의 설치하라.

```
linux@localhost:~$ sudo apt update
[sudo] password for linux:
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
linux@localhost:~$ sudo apt install gcc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  htop libnl-3-200 libnl-genl-3-200
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  cpp cpp-11 gcc-11 gcc-11-base libasan6 libatomic1 libc-dev-bin libc-devtools libc6-dev libcc1-0 libcryp
t-dev libgcc-11-dev libgd3 libgomp1 libisl23 libitm1 liblsan0 libmpc3
```

```
linux@localhost:~$ dpkg -l gcc
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name          Version          Architecture Description
+++-=====
ii gcc            4:11.2.0-1ubuntu1 amd64        GNU C compiler
linux@localhost:~$
```

□ 패키지 gcc의 삭제하라.

```
linux@localhost:~$ sudo apt remove gcc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  cpp cpp-11 gcc-11 gcc-11-base htop libasan6 libatomic1 libc-dev-bin libc-devtools libc6-dev libcc1-0
  libcrypt-dev libgcc-11-dev libgd3 libgomp1 libisl23 libitm1 liblsan0 libmpc3 libnl-3-200
  libnl-genl-3-200 libnsl-dev libquadmath0 libtirpc-dev libtsan0 libubsan1 libxpm4 linux-libc-dev
  manpages-dev rpcsvc-proto
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
```

```
linux@localhost:~$ dpkg -l gcc
dpkg-query: no packages found matching gcc
linux@localhost:~$
```